

Project Design Phase

Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	ShopEz (MERN Full-Stack Application)
Student Name	INAPAKUTHIKA BHARATH VARMA (bharathvarma906@gmail.com)
Maximum Marks	2 Marks

Proposed Solution Template:

The project team shall document the following details within the proposed solution framework.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Current e-retail architectures frequently encounter high request latency, inconsistent mobile responsiveness, and convoluted backend workflows. Such issues negatively impact user retention rates and cause operational inefficiencies in inventory control.
2.	Idea / Solution description	ShopEz employs the MERN stack to facilitate a high-velocity, single-page user experience. The application integrates instantaneous search capabilities, persistent shopping sessions, structured category filtering, and a consolidated administrative portal for holistic order and stock oversight.
3.	Novelty / Uniqueness	• Unified JavaScript environment (MERN) ensuring cohesive data synchronization. • Automated media optimization through Cloudinary integration. • Adaptive UI architecture tailored specifically for Men, Women, and Kids storefronts. • Advanced administrative modules for comprehensive lifecycle tracking of orders.

S.No.	Parameter	Description
4.	Social Impact / Customer Satisfaction	Improves the digital shopping landscape by providing a fortified and streamlined user journey. Increases consumer loyalty through rapid checkout procedures, real-time stock transparency, and intuitive access to transaction records.
5.	Business Model (Revenue Model)	• Direct retail margins and per-transaction processing fees. • Premium subscription tiers for third-party merchant toolsets. • Prioritized product visibility through promotional listing services.
6.	Scalability of the Solution	The NoSQL foundation of MongoDB facilitates seamless horizontal expansion of datasets. Future iterations are set to include multi-merchant integration, AI-powered suggestion engines, and support for global commerce via multi-currency payment systems.